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Theory of computation

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ASSIGNMENT-02

Task 1: Regular Expression Design

1. Binary strings ending with 01

Regular Expression: ^[01]*01$

Explanation: Accepts all binary strings that end with 01

2. Binary strings containing substring 11 or 00

RE: ^[01]*(11|00)[01]*$

Explanation: Accept binary strings containing 11 or 00

3. Strings over containing at least one occurance of ab

RE: ^[ab]*ab[ab]*$

Explanation: Accepts strings containing the substring ab at least once

4. Strings over 1 having an even number of zeros

RE: ^1*(01*01*)*$

Explanation: Accepts binary strings containing an even no. of zeros, including zero zero's

5. Strings over starting with a and ending with b

RE: ^a[ab]*b$

Explanation: Accepts strings that starts with a end with b.

Task 2- Regular Expression Analysis and Language Representation.

Given Regular Expression :

$$R = (0+1)^* \; 101 \; (0+1)^*$$

1. Language represented.

The regular expression represents all binary strings containing the substring 101 at least once

2. Alphabet and Substring

⇒ Alphabet : $\Sigma = 1$

⇒ Substring recognized : 101

3. five accepted strings.

1. 101
2. 0101
3. 1010
4. 11010
5. 10101

4. Three rejected strings

1. 000
2. 111
3. 1001

5. Kleene Star and union

Kleene Star (*) : Represents zero or more occurances of 0 or 1 represents and after The substring 101

Thus, (0+1) accepts either symbol

Conclusion: The expression accepts all binary strings containing 101 as a substring.

Task 3 - Context free Grammar (corrected)

1. $L_1 = \{a^n b^n \mid n \geq 0\}$

⇒ Start symbol : S

⇒ Terminals : a, b

Non Terminals

production rules

$S \rightarrow asb \mid \varepsilon$

2. $L_2 = \{a^n b^m \mid n, m \geq 0\}$

Start symbol : S

production rules

$S \rightarrow aS \mid A$

$A \rightarrow bA \mid \varepsilon$

4. Balanced paranthesis

Start symbol : S

Terminal symbols

NON-Terminal symbols

production rules :

$S \rightarrow (S)S \mid \varepsilon$

3. Airthematic expression involving identifiers and addition (+)

Start Symbol = E

Production rules:

$E \rightarrow E+E \mid id$.

5. Binary strings containing equal

Start Symbol : S

Terminal Symbols : 1

Production rules:

$S \rightarrow 0S1S \mid 1S0S \mid \varepsilon$

Task 4 - Derivation and parse Tree Construction

Given CFG

$S \rightarrow aSb \mid \varepsilon$

1. Components of grammar

production rules

$S \rightarrow aSb, \quad S \rightarrow \varepsilon$

Start symbol : S

2. Generate strings for $n = 0, 1, 2, 3$

n ⟶ String

0 ⟶ ε

1 ⟶ ab

2 ⟶ aabb

3 ⟶ aabb.

3. Leftmost derivation of aaabbb.

$S \rightarrow aSb$

$S \rightarrow asb$

$\rightarrow aaSbb$

$\rightarrow aaaSbbb$

$\rightarrow aaabbb$

Rightmost derivation of aaabbb

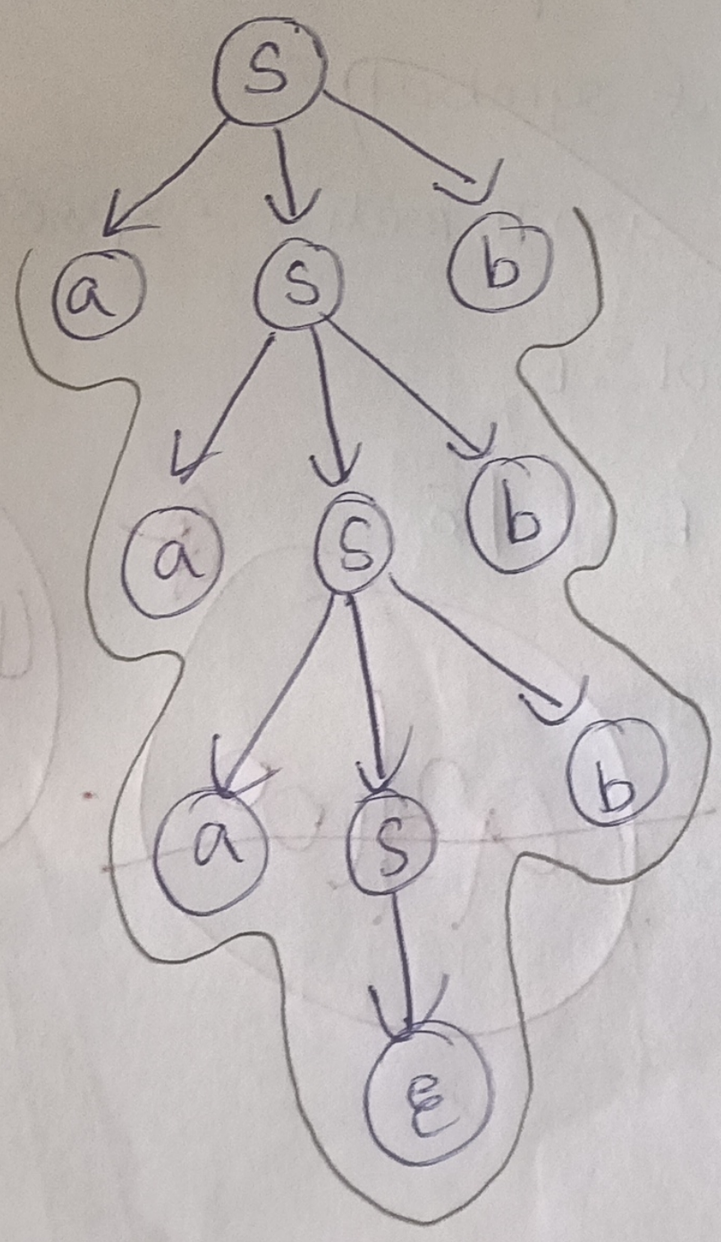
$S \Rightarrow asb$

$\Rightarrow aaSbb$

$\Rightarrow aaasbbb$

$\Rightarrow aaabbb$

parse tree for aaabbb



Task 5 :

Aim :

To construct a context free Grammar for arithmetic expressions containing identifiers, addition, multiplication, and parantheses while maintainies proper operator precedence.

2. Context - free Grammar

A CFG is represented as :

$$G_1 = (V, T, P, S)$$

where :

V : Set of non - terminals

T : Set of terminals

P : Set of production rules

S : Start symbol

for the given arithmetic expressions.

→ start symbol : E

$$E \rightarrow E+E \mid E$$